连续 / 准连续 $\xrightarrow{\text{调制不稳定}}$ 分裂为超短脉冲.

传输方程(无损耗).

$$i\frac{\partial A}{\partial z} = \frac{\beta_2}{2}\frac{\partial^2 A}{\partial T^2} - \gamma|A|^2 A.$$

稳态解:

$$\bar{A} = \sqrt{P_0}\exp(i\phi_{NL})$$

P_0: 入射功率

ϕ_{NL}: SPM 非线性相移.

引入微扰:

$$A = (\sqrt{P_0} + a)\exp(i\phi_{NL}).$$

代入并使 a 线性化,

$$i\frac{\partial a}{\partial z} = \frac{\beta_2}{2}\frac{\partial^2 a}{\partial T^2} - \gamma P_0(a + a^*).$$

由于 a^* 项, 频率 Ω 与 $-\Omega$ 分量发生耦合.

$$a(z,T) = a_1\exp[i(Kz - \Omega T)] + a_2\exp[-i(Kz - \Omega t)].$$

K、Ω 分别为微扰的波数与频率. 二者满足:

$$K = \pm\frac{1}{2}\left[|\beta_2\Omega|\Omega^2 + \mathrm{sgn}(\beta_2)\Omega_c^2\right]^{1/2} \qquad (5.1.6)$$

时才有一非平凡解.

$$\Omega_c^2 = \frac{4\gamma P_0}{|\beta_2|} = \frac{4}{|\beta_2|L_{NL}}. \qquad L_{NL} = \frac{1}{\gamma P_0}$$

由于 $e^{i(\beta_0 z - \omega_0 t)}$ 已提取出来, 实际微扰的波数与频率应为:

$\beta_0 \pm K$ 与 $\omega_0 \pm \Omega$.

由 5.1.6.

$\beta_2 > 0$. $K \in R$, 稳态在受微扰时稳定.

$\beta_2 < 0$. when $|\Omega| < \Omega_c$, $K \in I$. $K = K_r + iK_i$

$$a(z,t) = a_1 e^{i[(K_r+iK_i)z-\Omega T]} + a_2 e^{-i[(K_r+iK_i)z-\Omega T]}$$

$$= a_1 e^{-K_i z} e^{i(K_r z-\Omega T)} + a_2 e^{K_i z} e^{-i(K_r z-\Omega T)}$$

指数增加, 调制不稳定.